

YELLOWKNIFE, NT, Canada

WMO: 719360

Lat: 62.460N

Lon: 114.440W

Elev: 206

StdP: 98.88

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -40.0	(c) -37.7	(d) -44.2	(e) 0.0	(f) -39.7	(g) -42.4	(h) 0.1	(i) -37.5	(j) 9.2	(k) -19.5	(l) 8.3	(m) -20.3	(n) 1.8	(o) 310	(p) 0.757

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.0	25.6	16.1	23.9	15.3	22.3	14.6	17.2	22.9	16.4	21.9	15.6	20.9	4.1	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.1	11.0	19.0	14.1	10.3	18.5	13.1	9.6	18.1	49.4	22.9	46.8	21.9	44.4	21.0	20.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.1	8.2	7.3	-41.8	28.5	2.0	1.6	-43.2	29.7	-44.4	30.6	-45.5	31.5	-47.0	32.6
			-41.9	18.6	2.0	1.1	-43.3	19.4	-44.4	20.1	-45.5	20.7	-46.9	21.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-3.6	-25.0	-22.1	-16.4	-5.4	5.3	13.9	17.2	14.4	8.0	-0.8	-12.3	-21.3		
	DBStd	16.08	7.95	7.17	8.27	7.27	5.59	4.05	3.13	3.52	4.29	4.60	6.91	8.20		
	HDD10.0	5510	1084	899	819	463	163	12	0	6	88	334	670	972		
	HDD18.3	8051	1343	1133	1077	713	404	139	59	129	311	593	920	1230		
	CDD10.0	540	0	0	0	0	17	130	223	142	27	1	0	0		
	CDD18.3	38	0	0	0	0	0	8	24	7	0	0	0	0		
	CDH23.3	198	0	0	0	0	1	48	113	37	0	0	0	0		
	CDH26.7	19	0	0	0	0	0	4	11	4	0	0	0	0		
(14) Wind	WSAvg	3.5	2.9	3.0	3.4	3.7	3.9	3.6	3.4	3.5	3.8	3.7	3.5	3.1		
(15) Precipitation	PrecAvg	287	15	14	13	11	18	26	35	42	32	39	26	16		
	PrecMax	438	34	36	40	27	62	73	110	146	64	103	46	47		
	PrecMin	179	0	2	2	1	0	2	6	4	6	10	8	5		
	PrecStd	53	8	8	7	7	14	18	24	33	15	19	10	8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-3.3	-3.3	3.7	12.9	22.4	27.0	28.2	27.1	20.3	12.6	2.3	-2.8		
		MCWB	-4.4	-4.6	0.6	6.4	12.0	16.5	17.2	17.1	13.8	8.9	1.2	-3.6		
	2%	DB	-7.6	-7.0	0.9	9.2	19.1	24.7	26.0	23.9	17.6	8.7	0.4	-4.9		
		MCWB	-8.1	-8.0	-1.3	4.3	10.4	15.0	16.5	15.6	12.6	6.3	-0.4	-5.5		
	5%	DB	-11.3	-9.5	-1.4	6.9	16.3	22.9	24.4	22.0	15.7	6.7	-1.5	-7.4		
		MCWB	-11.8	-10.5	-3.2	2.9	9.1	14.0	15.9	15.1	11.6	4.9	-2.2	-7.8		
	10%	DB	-14.0	-11.7	-3.9	4.7	14.0	21.1	22.9	20.1	14.0	5.0	-3.3	-10.1		
		MCWB	-14.6	-12.5	-5.4	1.5	7.7	13.3	15.2	14.5	10.8	3.6	-3.9	-10.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-4.3	-4.6	1.0	6.7	12.7	17.4	18.7	18.2	14.9	9.4	1.4	-3.4		
		MCDB	-3.3	-3.3	3.4	12.3	20.6	24.7	25.0	23.6	18.8	12.4	2.4	-2.9		
	2%	WB	-8.2	-7.8	-1.2	4.6	10.9	16.0	17.6	16.8	13.2	6.7	-0.3	-5.5		
		MCDB	-7.6	-7.1	0.9	8.8	18.4	22.7	23.4	21.6	16.4	8.5	0.3	-4.9		
	5%	WB	-11.8	-10.4	-3.2	3.0	9.5	15.0	16.8	15.9	12.1	5.0	-2.1	-7.8		
		MCDB	-11.3	-9.6	-1.3	6.5	15.8	21.2	22.4	20.4	15.2	6.6	-1.6	-7.4		
	10%	WB	-14.6	-12.5	-5.4	1.6	8.0	13.9	16.0	15.0	11.0	3.7	-3.9	-10.5		
		MCDB	-14.0	-11.7	-3.9	4.5	13.6	20.0	21.7	19.3	13.7	5.0	-3.3	-10.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.0	8.5	10.8	10.9	9.7	8.9	8.0	7.1	6.1	4.6	6.5	7.0		
		MCDBR	7.6	8.5	10.5	11.0	12.5	11.2	10.0	9.8	8.6	6.3	6.2	8.1		
		MCWBR	7.4	8.1	8.8	7.4	6.8	5.1	4.0	4.2	4.5	4.6	6.0	8.0		
		MCDBR	7.6	8.5	10.5	10.3	12.0	10.1	8.9	8.5	8.1	6.0	6.3	8.0		
	5% WB	MCWBR	7.5	8.1	8.9	7.0	6.8	5.0	4.2	4.4	4.6	4.6	6.1	7.9		
(40) Clear-Sky Solar Irradiance	taub		0.193	0.243	0.270	0.305	0.326	0.345	0.369	0.339	0.310	0.275	0.192	0.156		
	taud		2.030	2.207	2.234	2.275	2.462	2.447	2.418	2.518	2.558	2.411	2.042	1.931		
	Ebn at Noon		502	733	845	883	894	879	844	845	810	694	488	350		
	Edh at Noon		33	61	87	105	96	100	100	83	66	51	31	18		
(44) All-Sky Solar Radiation	RadAvg		0.26	1.04	2.79	4.85	5.91	6.20	5.47	4.09	2.47	0.95	0.28	0.11		
	RadStd		0.03	0.09	0.21	0.26	0.37	0.53	0.29	0.20	0.21	0.11	0.04	0.01		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	+1.43	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page